COMSATS University Islamabad, Lahore Campus

Terminal Lab Examination – Fall 2025

Course Title:	Web Technologies			Course Code:	CSC336	Credit	3(2,1)
Course Instructor/s:	Asif Shahzad			Programme Name:	BSCS		
Semester:	5	Batch:	FA23	Section:	G1	Date:	Dec. 18, 2025
Time Allowed:	**1.5 Hours**			**Maximum Marks:**	**50**		
Student's Name:				Reg. No.			

Important Instructions

1. Store data in `patients.json` and `visits.json` files **OR** use any online database.
2. I will open **only the home page** using: `localhost:3000/`. All other pages must be properly linked and must not require entering URLs manually.
3. Evaluation will be done with **viva**.
4. After completion, make a copy of your project and **delete only the `node_modules` folder**. Do not remove any other file. There must be **at least 2 patients already added** when you upload.
5. Zip file name format: `<registration-number>-<your-name>.zip`

Question 1: ***CLO 5***

Clinic Token & Visit Management System

Create a web application for a **doctor's clinic** where patients are served using **token numbers**.

<u>Patient & Token Management (15 Marks)</u>

- Implement **Add, Edit, and Remove Patient**. Patient fields: full name, age, contact number.
- Generate a **daily token number** automatically when a patient is added.
- Display today's token list on the home page (/) along with patient information in tabular form.

<u>Visit Handling & Earnings (15 Marks)</u>

- Doctor can **call a token** by clicking a "Call" button, displayed at home page against each patient
- When "Call" button is clicked, display two options to choose whether patient actually came or not:
  - **Mark as Visited**
  - **Mark as No-Show**
- If a patient is called and status is saved, that row should not display "Call" button
- Store visit status in `visits.json`.
- Use **AJAX** for calling token and marking visit status.

- At home page, also display following information as summary in one row with proper labels:
  - Total tokens issued today, total patients added today
  - Total patients who actually visited
  - Total patients who did not show up
  - **Total fee earned for the day**
- Assume **doctor fee = 1000 per visited patient**.

- Earnings must be calculated dynamically based on visit data.

Authentication (15 Marks)

- Provide a **Doctor Login**.
- Only logged-in doctor can:
  - Call tokens
  - Mark visit status
  - View daily earnings
- Fixed credentials: `doctor / 123`.

UI / UX (5 Marks)

- Clear and simple interface suitable for clinic use.
- Token call action and status updates must not reload the page.